

# Guide to run server

## 1. Introduction

This document outlines the steps required to set up and run the backend service in a local development environment.

The backend is implemented using **AWS Lambda**, **Terraform**, **Docker**, and **Dev Containers**.

Following this guide ensures a consistent and reliable environment for all team members.

## 2. Prerequisites

### 2.1 Docker

To support containerized local execution:

1. Download and install **Docker Desktop**:
2. <https://www.docker.com/products/docker-desktop>
3. Complete the installation according to your operating system.
4. Ensure that the Docker Desktop is running.
5. Verify the installation:

```
docker --version
```

### 2.2 Visual Studio Code

1. Install **Visual Studio Code**:

<https://code.visualstudio.com/>

2. Install the **Dev Containers** extension:

<https://marketplace.visualstudio.com/items?itemName=ms-vscode-remote.remote-containers>

## 2.3 LocalStack

LocalStack is used to emulate AWS services locally.

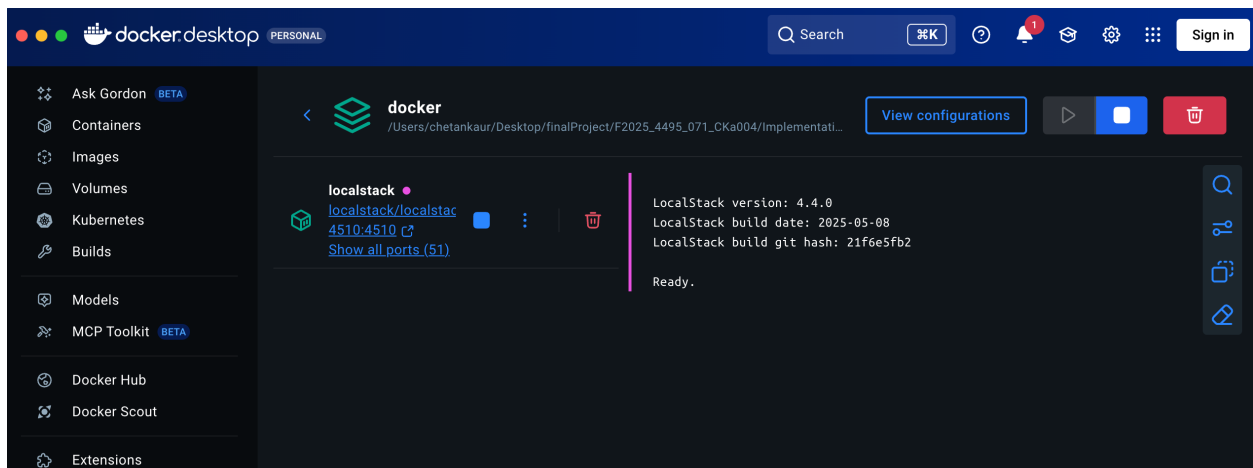
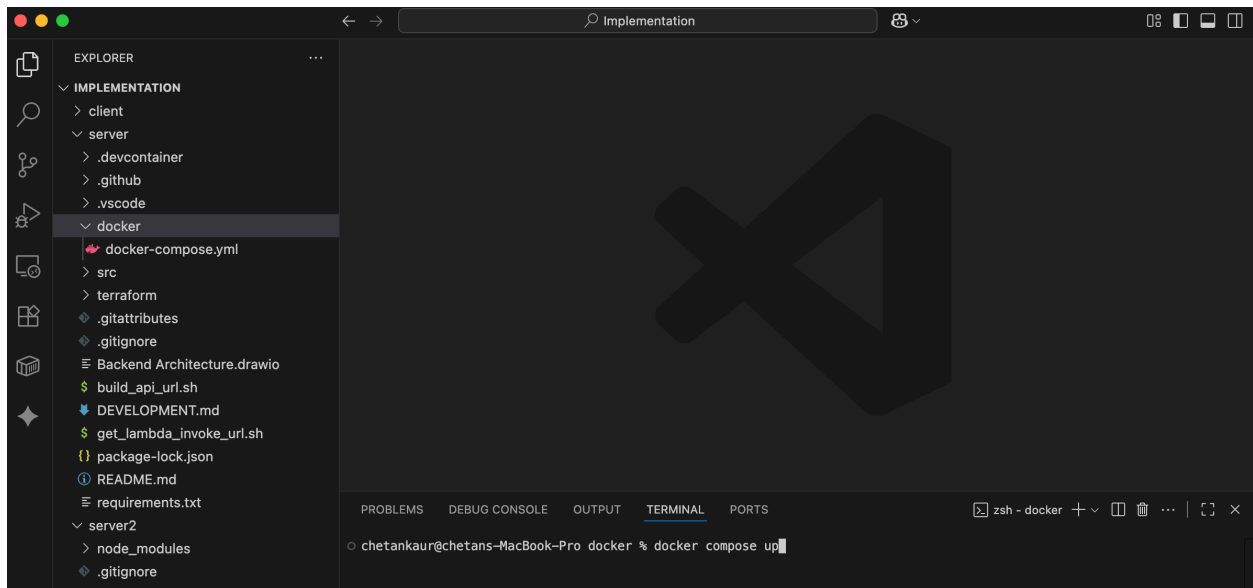
- Access the LocalStack web interface:

<https://app.localstack.cloud/inst/default/status>

# 3. Running the Backend Locally

## 3.1 Start Docker Services

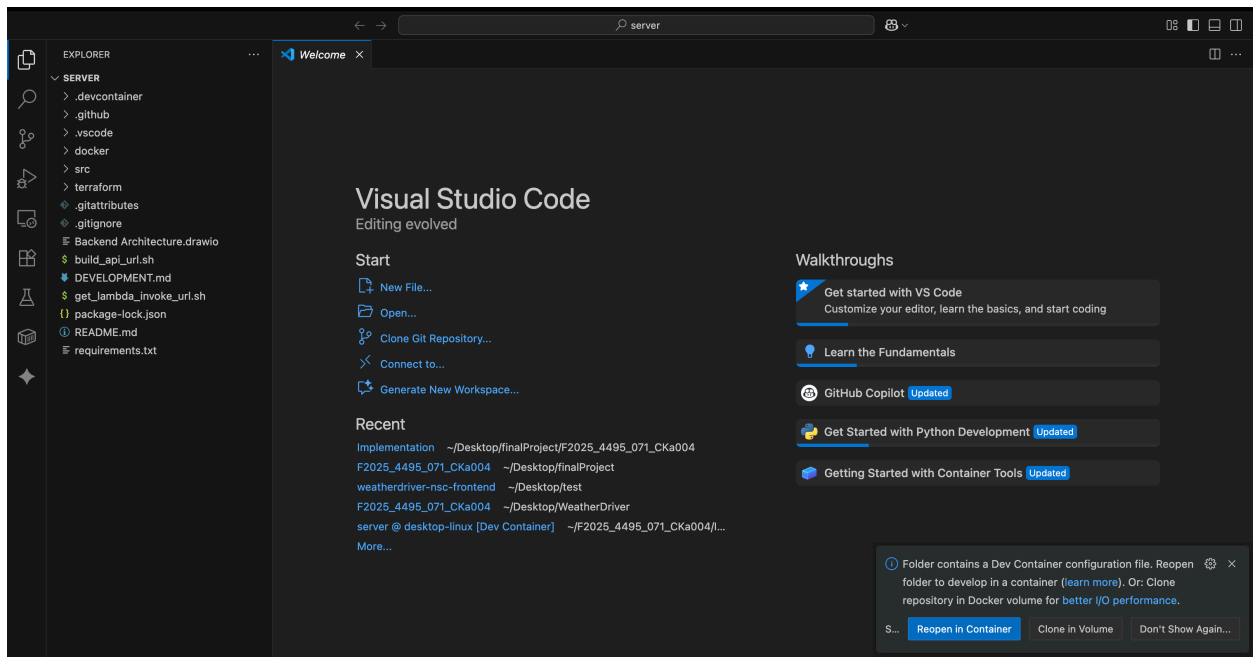
1. Open the repository in **Visual Studio Code**.
2. Navigate to the docker directory.
3. Start the required Docker services: **docker compose up**



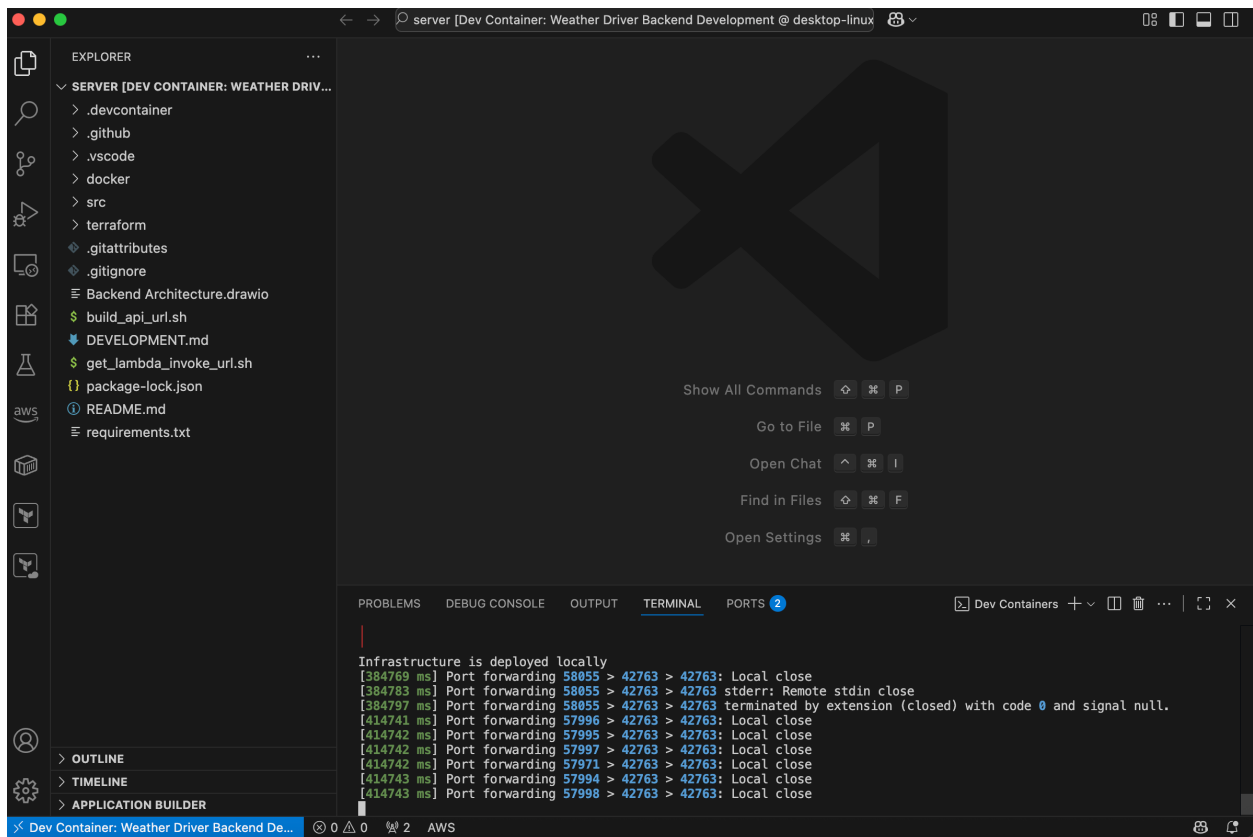
## 3.2 Launch the Development Container

To ensure a consistent development environment:

1. Open a **new instance** of Visual Studio Code.
2. Open the repository in this new instance.
3. When prompted or via the Command Palette (Ctrl+Shift+P / Cmd+Shift+P)
4. select: Dev Containers: Reopen in Container



Once the project is reopened in the development container, all tools and dependencies required for backend development will be automatically provisioned.



## 4. Running Lambda Functions Locally

AWS Lambda functions can be executed locally using **LocalStack** and the **awslocal** CLI.

This allows developers to test Lambda behavior, event payloads, and S3/SQS integrations without deploying to AWS.

### 4.1 Listing Available Lambda Functions

To verify that all Lambda functions have been deployed to LocalStack, run:

```
awslocal lambda list-functions
```

This returns metadata including runtime, memory size, timeout, and log configuration.

### 4.2 Invoking a Lambda Function With a JSON Payload

Lambda functions may be invoked with inline JSON or by using a JSON file.

#### Invoke using a local payload file

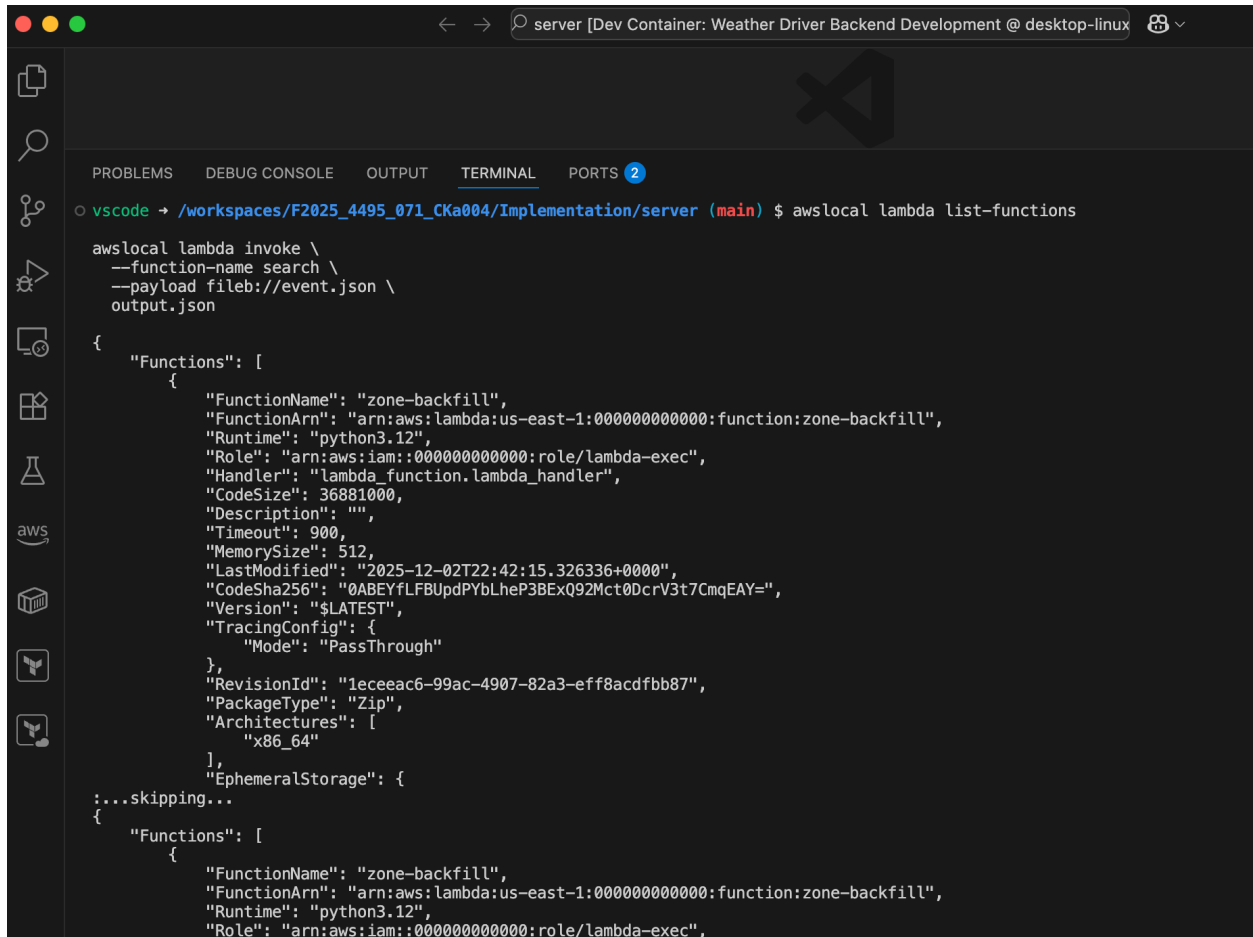
Recommended approach :

```
awslocal lambda invoke \ --function-name <function-name> \ --payload fileb://<path-to-event-  
file>.json \  
    output.json
```

Example from your environment:

```
awslocal lambda invoke \ --function-name search \ --payload fileb://event.json \output.json
```

View the response: `cat output.json`



```
server [Dev Container: Weather Driver Backend Development @ desktop-linux]
PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS 2
vscode → /workspaces/F2025_4495_071_CKa004/Implementation/server (main) $ awslocal lambda list-functions

awslocal lambda invoke \
  --function-name search \
  --payload fileb://event.json \
  output.json

{
  "Functions": [
    {
      "FunctionName": "zone-backfill",
      "FunctionArn": "arn:aws:lambda:us-east-1:000000000000:function:zone-backfill",
      "Runtime": "python3.12",
      "Role": "arn:aws:iam:000000000000:role/lambda-exec",
      "Handler": "lambda_function.lambda_handler",
      "CodeSize": 36881000,
      "Description": "",
      "Timeout": 900,
      "MemorySize": 512,
      "LastModified": "2025-12-02T22:42:15.326336+0000",
      "CodeSha256": "0ABEYfLFBUpdPYbLheP3BExQ92Mct0DcrV3t7CmqEAY=",
      "Version": "$LATEST",
      "TracingConfig": {
        "Mode": "PassThrough"
      },
      "RevisionId": "1eceeac6-99ac-4907-82a3-ef8acdfbb87",
      "PackageType": "Zip",
      "Architectures": [
        "x86_64"
      ],
      "EphemeralStorage": {
        ...skipping...
      }
    }
  ]
}
```

## 4.3 Invoking Lambda Functions That Consume SQS Events

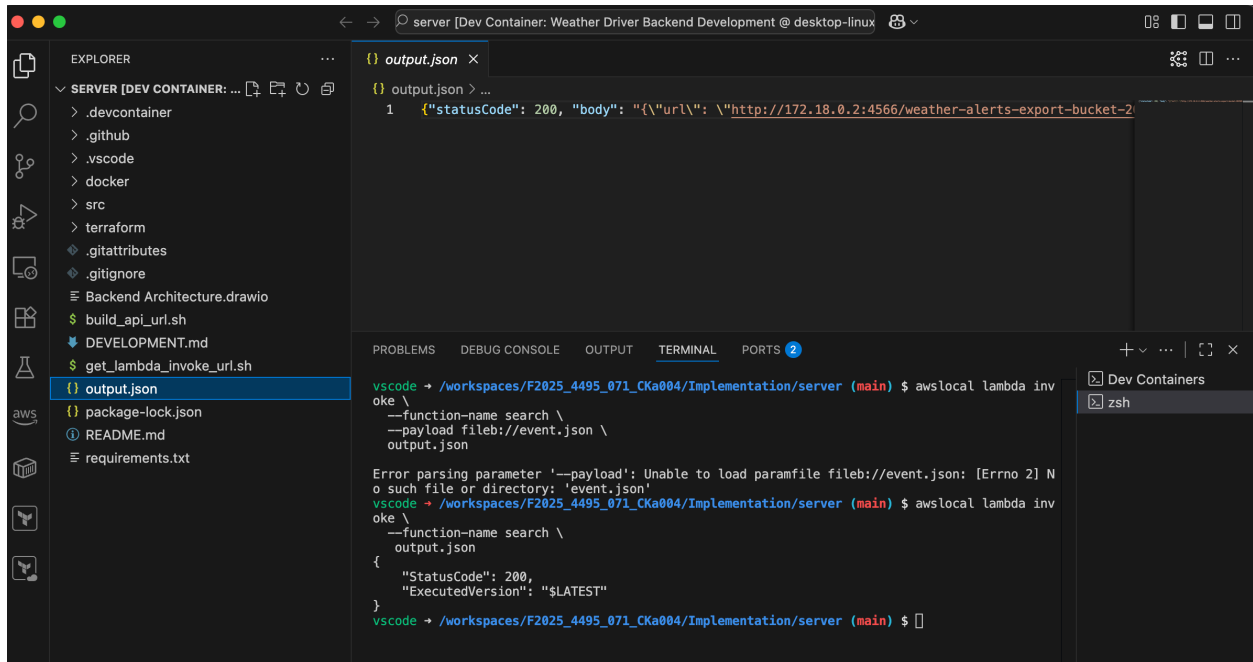
For Lambdas triggered by SQS, pass a mock SQS event file structured as "Records": [...].

Example:

```
awslocal lambda invoke \ --function-name alert-listener \ --payload
fileb://src/aws/lambda/alert_listener/sqs.json \ output.json
```

Running the search lambda function and this will link to download the output in output.json

```
awslocal lambda invoke --function-name search output.json
```



## 4.4 Invoking Lambda Functions That Process S3 Files

Some Lambdas use S3 input objects.

To test them:

### Step 1 — Copy the S3 test file from LocalStack to your container

```
awslocal s3 cp \  
  "s3://weather-alerts-export-bucket-202505/56220255-916f-4f24-a42d-fc46f2a8b6a4.json" \  
  ./event-999.json
```

```
{ } output.json X
{} output.json > body
1 /172.18.0.2:4566/weather-alerts-export-bucket-202505/2471eb38-1e28-4de2-8152-f9301347f37f.json?AWSAccessKeyId=LSIAQAAAAAADALNESB6&S

PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS 2
vscode + /workspaces/F2025_4495_071_CKa004/Implementation/server (main) $ awslocal lambda invoke \
--function-name search \
--payload fileb://event.json \
output.json
Error parsing parameter '--payload': Unable to load paramfile fileb://event.json: [Errno 2] No such file or directory: 'event.json'
vscode + /workspaces/F2025_4495_071_CKa004/Implementation/server (main) $ awslocal lambda invoke \
--function-name search \
output.json
{
  "StatusCode": 200,
  "ExecutedVersion": "$LATEST"
}
vscode + /workspaces/F2025_4495_071_CKa004/Implementation/server (main) $ awslocal s3 cp \
"s3://weather-alerts-export-bucket-202505/2471eb38-1e28-4de2-8152-f9301347f37f.json" \
./event-999.json
```

```
server [Dev Container: Weather Driver Backend Development @ desktop-linux]
EXPLORER
SERVER [DEV CONTAINER: WEATH...
  > .devcontainer
  > .github
  > .vscode
  > docker
  > src
  > terraform
  > .gitattributes
  > .gitignore
  > Backend Architecture.drawio
  > build_api_url.sh
  > DEVELOPMENT.md
  > event-999.json
  > get_lambda_invoke_url.sh
  > output.json
  > package-lock.json
  > README.md
  > requirements.txt

{} event-999.json X
{} event-999.json > {} 0 > title
1 [{"id":"urn:oid:2.49.0.1.840.0.1d3c31ff9022368436f681b86e7da4b0d882ac95.005.1",
"start":"2025-12-02T12:25:00-07:00","end":"2025-12-02T23:00:00-07:00","updated":"2025-12-02T22:54:26.504809+00:00","severity":"Moderate","event":"Expected","title":"Winter Weather Advisory issued December 2 at 12:25PM MST until December 2 at 11:00PM MST by NWS Great Falls MT","message":"* WHAT...Snow. Additional snow accumulations up to two inches.\n\n* WHERE...Northwest Beaverhead County.\n\n* WHEN...Until 11 PM MST this evening.\n\n* IMPACTS...Plan on slippery road conditions.",
"link":"https://api.weather.gov/alerts/urn:oid:2.49.0.1.840.0.1d3c31ff9022368436f681b86e7da4b0d882ac95.005.1","min_lat":0.00000,"max_lat":0.00000,"min_lon":0.00000,"max_lon":0.00000,"geometry":[],"id":"urn:oid:2.49.0.1.840.0.9a5af99817b4d9edcfd61f0327d754f8de44e19.003.1","start":"2025-12-02T18:17:00-08:00","end":"2025-12-03T09:00:00-08:00","updated":"2025-12-03T07:14:50.463327+00:00","severity":"Minor","event":"Expected","title":"Frost Advisory issued December 2 at 6:17PM PST until December 3 at

PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS 2
vscode + /workspaces/F2025_4495_071_CKa004/Implementation/server (main) $ awslocal s3 cp "s3://weather-alerts-export-buc
ket-202505/2471eb38-1e28-4de2-8152-f9301347f37f.json" event-999.json
download: s3://weather-alerts-export-bucket-202505/2471eb38-1e28-4de2-8152-f9301347f37f.json to ./event-999.json
vscode + /workspaces/F2025_4495_071_CKa004/Implementation/server (main) $
```